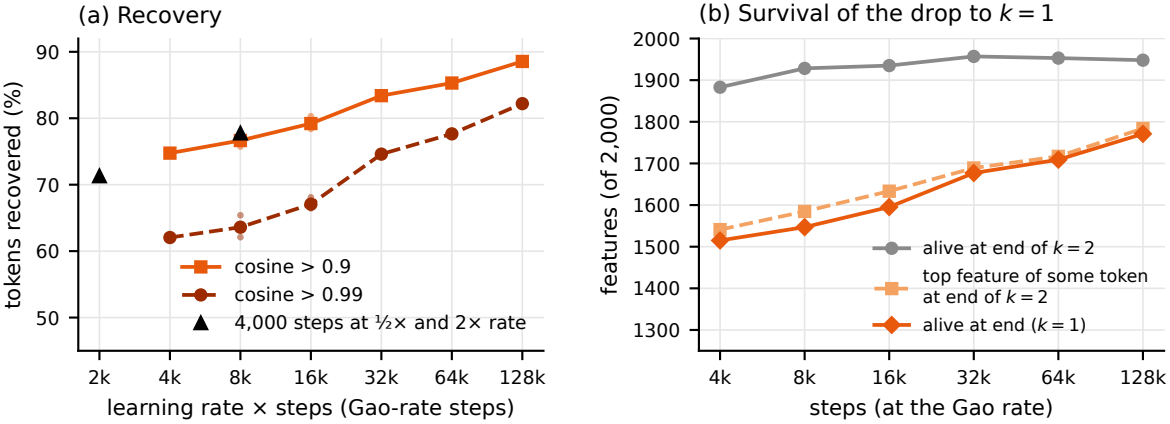
(a) Recovery
tokens recovered (%)
90
80
70
60
50
cosine > 0.9
cosine > 0.99
4,000 steps at ½× and 2× rate
2k
4k
8k
16k
32k
64k
128k
learning rate × steps (Gao-rate steps)
(b) Survival of the drop to $k = 1$
features (of 2,000)
2000
1900
1800
1700
1600
1500
1400
1300
alive at end of $k = 2$
top feature of some token at end of $k = 2$
alive at end ($k = 1$)
4k
8k
16k
32k
64k
128k
steps (at the Gao rate)